

CSC 365 Lab 13 JS Loops and Conditionals

In this lab, you will create a rock, paper, scissors game that the user plays against the computer. The scores will be counted, and the best two out of three wins. Do not play the third round if one player gets 2 points before the other gets any. The user should enter “r” for rock, “p” for paper, or “s” for scissors. The computer will generate a random number which will correspond to the computer’s play. The rules are no points for a tie, paper covers rock, rock crushes scissors, scissors cut paper:



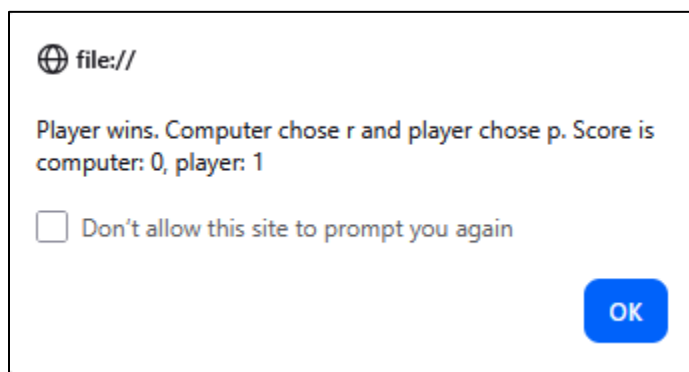
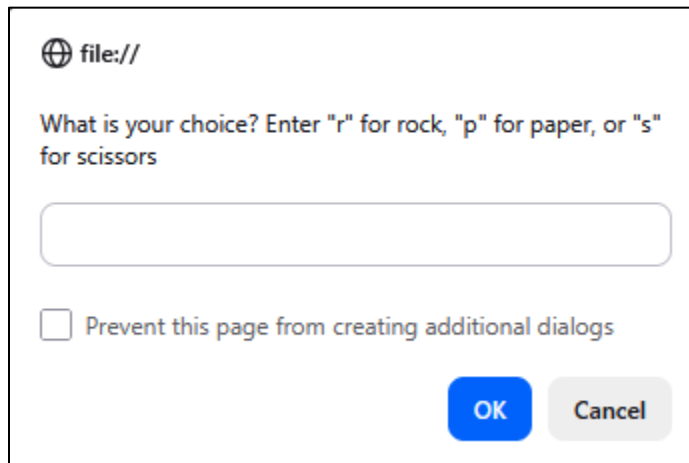
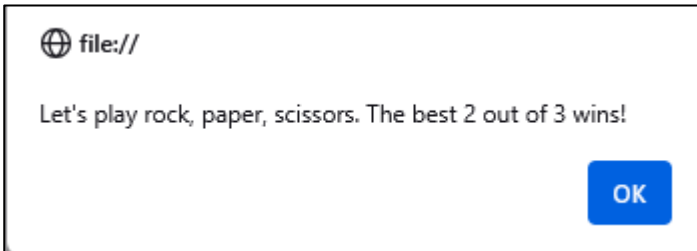
The Math method `Math.random()` will pick a number between 0 and 1. To pick a number between 0 and 2 (inclusive), multiply that number by 3 and then apply the `Math.floor()` method to get an integer between 0 and 2 inclusive.

1. Download and open `rps.html`
2. Add your name to the comment on line 2
3. Use **const** to create an array to store the characters “r”, “p”, and “s”.
4. Create and initialize two integers to store each player’s total score.
5. Create two variables to store the user’s choice (a string) and the computer’s choice (also a string)
6. Create another variable to store a random number.
7. Use a while loop to control when the game ends.
8. In the while loop do the following:
 - a. Prompt the player for their choice (“r” for rock, “p” for paper, or “s” for scissors.
 - b. Use the method `toLowerCase()` to convert the user’s entry to lower case.
 - c. Use a loop to force the user to enter a valid character.
 - d. Generate the random number 1, 2, or 3.
 - e. Determine the computer’s play as determined by the random number used. For example, if the random number is 0, the computer’s play is rock because 0 is the first element of the array.
 - f. Determine if the round is a tie, otherwise determine the winner and increment the winner’s score.
 - g. Output who won the round using an alert statement that displays what each player chose and the new scores.

- h. At the end of the game, declare the winner
 - i. See screenshots below for examples.
9. Look at your code. Do you see redundancy? If so, refactor so that the code is as concise as possible.
10. Submit your html file into Canvas.

Some example screenshots of a game

At the beginning:



Later:

 file://

It's a tie. Computer chose s and player chose s. Score is computer: 1, player: 1

☐ Don't allow this site to prompt you again

OK

At the end:

 file://

Game over. Final score is computer: 1, player: 2

☐ Don't allow this site to prompt you again

OK